

CS 331 (Software Engineering Lab)

Assignment 7 (Total Marks = 40)

A **"business logic layer" (BLL)** is a part of a software application architecture where the core business rules and logic are implemented, acting as a mediator between the presentation layer (user interface) and the data access layer (database), ensuring that data manipulation and processing adhere to the specific business requirements and guidelines of the application.

Q1. Identify the core functional modules related to the business logic layer of your software. Implement them and clearly show their interaction with the components of the presentation layer (user interface) that you have already implemented as part of your previous lab assignment.

[Marks = 10]

Q2. Describe the following for your software engineering project. **[Marks = 10x3 = 30]**

- A) How are the business rules implemented for different modules of your project? [Business rules refer to the rules and conditions an application follows to perform various operations, like validating whether a user can access a resource, applying discount rules on products, etc.]
- B) Validation logic ensures that data entering the system is correct, consistent, and in the proper format before being processed further. Have you implemented any validation logic for your application? Explain.
- C) Data Transformation: Data coming from the data layer (usually databases) may need to be transformed into a format that can be used by the presentation layer, which is the UI. How have you taken care of proper data transformation for your software project?